Setting up a Notesnook Server on a Raspberry Pi 5

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Setup a Notesnook server on a Raspberry Pi 5 (64-bit recommended) using a mix of sideloaded and official services, integrated with Cloudflare Tunnel and Tailscale for secure access.

> **Note:** This guide uses a Cloudflare Tunnel for frontend access, but the sensitive sync/attachment traffic bypasses it entirely through Tailscale for simplicity.

Prerequisites

- Raspberry Pi 5 with 64-bit OS installed, SSH access configured
- Cloudflare account with a domain you own
- Tailscale account (free tier is fine)

Architecture Overview

```
Internet
   ├── Cloudflare Tunnel ── notes.yourdomain.com ── :80 (web app)
   └── Tailscale ── :9090 (sync/attachments)

Docker Network (notesnook-net)
   ├── notesnook-web      (nginx, port 80)
   ├── notesnook-sync     (port 9090)
   ├── notesnook-attachments (port 3126)
   ├── mongo              (port 27017, internal only)
   ├── minio              (port 9000, internal only)
   ├── redis              (port 6379, internal only)
   ├── caddy              (port 443, internal only)
   ├── s3-init            (one-shot bucket creation)
   ├── cloudflared        (tunnel to Cloudflare)
   └── tailscale          (tailscale0 interface)
```

Setting up the Docker Stack

1. Create a folder structure for the MinIO and Caddy configurations (mkdir -p minio/data caddy/data caddy/config).
2. Create a .env file containing your secrets and the MongoDB replica set key (openssl rand -base64 756 > mongo.key).
3. Create the docker-compose.yaml file. Key points:
   - Use `bitnami/mongodb:latest` with `MONGODB_EXTRA_FLAGS=--wiredTigerCacheSizeGB=0.25` to limit memory.
   - Use `minio/minio:latest` and `minio/mc:latest` for S3-compatible storage.
   - Use `caddy:2-alpine` with a custom Caddyfile to proxy the attachments subdomain.
   - Use `cloudflare/cloudflared:latest` for the tunnel.
   - Use `tailscale/tailscale:latest` with `network_mode: host` for Tailscale.
   - Use `redis:7-alpine`.
   - Use `nginx:alpine` for the web app.
   - Use `node:20-alpine` for the sync and attachments services.

```
services:
  s3-init:
    image: minio/mc:latest
    container_name: s3-init
    networks: [notesnook-net]
    depends_on:
      minio: { condition: service_healthy }
    entrypoint: >-
      /bin/sh -c
      "
      mc alias set s3 http://minio:9000 $$MINIO_ROOT_USER$$ $$MINIO_ROOT_PASSWORD$$ --api s3v4;
      mc mb --ignore-existing s3/attachments;
      mc anonymous set download s3/attachments/avatars;
      exit 0;
      "
    environment:
      MINIO_ROOT_USER: ${MINIO_ROOT_USER}
      MINIO_ROOT_PASSWORD: ${MINIO_ROOT_PASSWORD}
```

4. Run `docker compose up -d`.
5. Check logs with `docker compose logs -f`.

Cloudflare and Tailscale Setup

- Create a tunnel in Cloudflare Zero Trust dashboard.
- Add a public hostname: `notes.yourdomain.com` → `http://localhost:80`.
- Install Tailscale on the host: `curl -fsSL https://tailscale.com/install.sh | sudo sh`.
- Run `sudo tailscale up --accept-routes --accept-dns=false`.
- Note the Tailscale IP (`tailscale ip -4`).

Setup on Cloudflare Tunnel

Cloudflare's tunnel is perfect where there is zero or sufficient ports available and port forwarded. Since we're using Tailscale for sync and attachments, we only need the web app exposed through the tunnel. Cloudflare handles HTTPS, DDoS protection, and hides your home IP.

1. Create a tunnel in Cloudflare Zero Trust dashboard.
2. Add a public hostname: `notes.yourdomain.com` → `http://localhost:80`.
3. Copy the tunnel token.
4. Add `TUNNEL_TOKEN=...` to your `.env`.
5. Add the `cloudflared` service to docker-compose:

```
  cloudflared:
    image: cloudflare/cloudflared:latest
    container_name: cloudflared
    networks: [notesnook-net]
    restart: unless-stopped
    command: tunnel --no-autoupdate run --token ${TUNNEL_TOKEN:-dummy}
```

6. Run `docker compose up -d cloudflared`.
7. Verify tunnel connection in Cloudflare dashboard.

Setup on Tailscale

1. Install Tailscale on the host: `curl -fsSL https://tailscale.com/install.sh | sudo sh`.
2. Run `sudo tailscale up --accept-routes --accept-dns=false`.
3. Note the Tailscale IP: `tailscale ip -4`.
4. Use this IP in `S3_INTERNAL_ENDPOINT` for attachments.
5. Access sync at `http://<tailscale-ip>:9090`.

Setup on Caddy

1. Create `caddy/Caddyfile`:

```
:80 {
    reverse_proxy minio:9000
}
```

2. Add `caddy:2-alpine` to docker-compose.
3. Attachments will be available at `http://<tailscale-ip>:3126`.

Setup on Notesnook Web App

1. Create `web.Dockerfile`:

```
FROM node:20-alpine AS build
RUN apk add git
RUN git clone --depth 1 https://github.com/streetwriters/notesnook.git /app
WORKER_URL="https://attachments.notes.yourdomain.com" \
WORKER_URL_SOCKET="wss://sync.notes.yourdomain.com/hub" \
MINIMUM_SERVER_VERSION="0.0.0" \
DISABLE_SIGNUP_BY_DEFAULT="false" \
DISABLE_SSO="false" \
DISABLE_MFA="false" \
DISABLE_RECOVERY_KEY="false" \
DISABLE_IMPORT="false" \
DISABLE_SUBSCRIPTIONS="true" \
DISABLE_ANALYTICS="true" \
DISABLE_ERROR_MONITOR="true" \
DISABLE_PAYMENT_PROVIDER="true" \
DISABLE_SSE="false" \
DISABLE_SYNC="false" \
DISABLE_ATTACHMENTS="false" \
DISABLE_MONETIZATION="true" \
DISABLE_TRIAL="false" \
DISABLE_SELF_HOSTED="true" \
DISABLE_ACCOUNT_CREATION="false" \
DISABLE_PASSWORD_RESET="false" \
DISABLE_EMAIL_CHANGE="false" \
DISABLE_USERNAME_CHANGE="false" \
DISABLE_PROFILE_PICTURE="false" \
DISABLE_TWO_FACTOR="false" \
DISABLE_SESSIONS="false" \
DISABLE_API_TOKENS="false" \
DISABLE_SSO_PROVIDERS="false" \
DISABLE_MFA_PROVIDERS="false" \
DISABLE_OAUTH="false" \
DISABLE_GOOGLE_DRIVE="true" \
DISABLE_DROPBOX="true" \
DISABLE_ONEDRIVE="true" \
DISABLE_NEXTCLOUD="true" \
DISABLE_WEBDAV="true" \
DISABLE_SMTP="true" \
DISABLE_IMAP="true" \
DISABLE_MAILGUN="true" \
DISABLE_PUSH
```